



7912 - A New Window on Galaxy Structure: Mid-IR Morphology with MEGA

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

In the local universe, the dustiest galaxies are universally mergers. Previous studies of optical morphologies indicate this is not necessarily the case at cosmic noon, but no study to date has directly examined the morphology of the dust. Optical morphologies are insufficient to tease out relationships in an obscured population. JWST has given the astronomical community access to sub-arcsecond resolution imaging well into the mid-IR, making these dust morphologies attainable for the first time. We propose a novel and comprehensive morphological study on mid-IR data from the MEGA survey, as well as archival data from the CEERS and CANDELS programs. We will produce and publish a value-added catalog of IR and optical morphological metrics, comprised of visual classifications from Zooniverse, Sérsic indices from GALFIT, bulge-to-light ratios and effective radii from GaMPEN, and CAS, GINI, and M20 measurements. We will also map the surface density of obscured star formation, stellar mass, and dust, for resolved objects at redshifts where PAH emission falls into a MIRI band. We will use this, coupled with global galaxy properties derived from SED fitting with CIGALE, to: describe how dust and stellar morphology evolves with cosmic time and varies with physical properties; to examine how AGN growth correlates with merger signatures in faint, high- z , and obscured populations; and to shed light on how and where star formation is quenched.

OBSERVING DESCRIPTION